March Report on FYP

Machine Learning Assisted Real-time Diagnosis on CO2 Discharge Process

Mingjie Zhu (Jay Zhu)

sgmzhu11@liverpool.ac.uk

University of Liverpool

March 9, 2026

Table of Contents

1. Project Overview
2. Phase 1: Baseline Modeling with PCA
3. Phase 2: Hyperparameter Tuning
4. Phase 3: Domain-Knowledge Feature Engineering
5. Phase 4: Interpretability & Feature Reduction
6. Summary

Problem Statement

- Nanosecond pulsed CO_2 discharge in water produces H_2O_2 — a key oxidant for water treatment
- Optical Emission Spectroscopy (OES) captures real-time plasma spectra (200–900 nm, 701 channels)
- **Goal:** Predict H_2O_2 yield rate from OES data and discharge parameters using machine learning
- **Challenge:** Only 20 experimental samples available (small-sample regime)

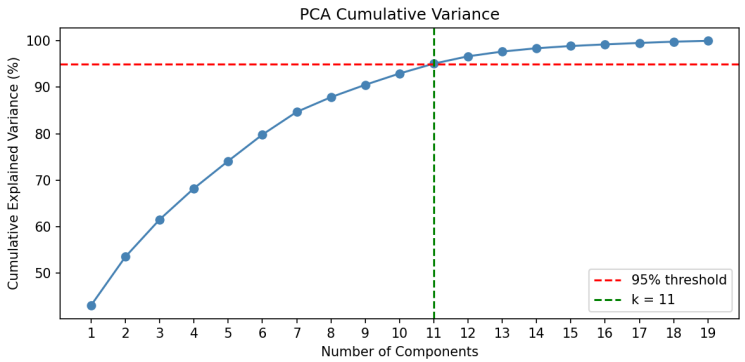
Experimental Data & Configurations

Config	Input Features	Dimensions
A	OES only (PCA-reduced)	11
B	Discharge parameters only	4
C	OES + discharge parameters	15 (P1) / 17 (P3)

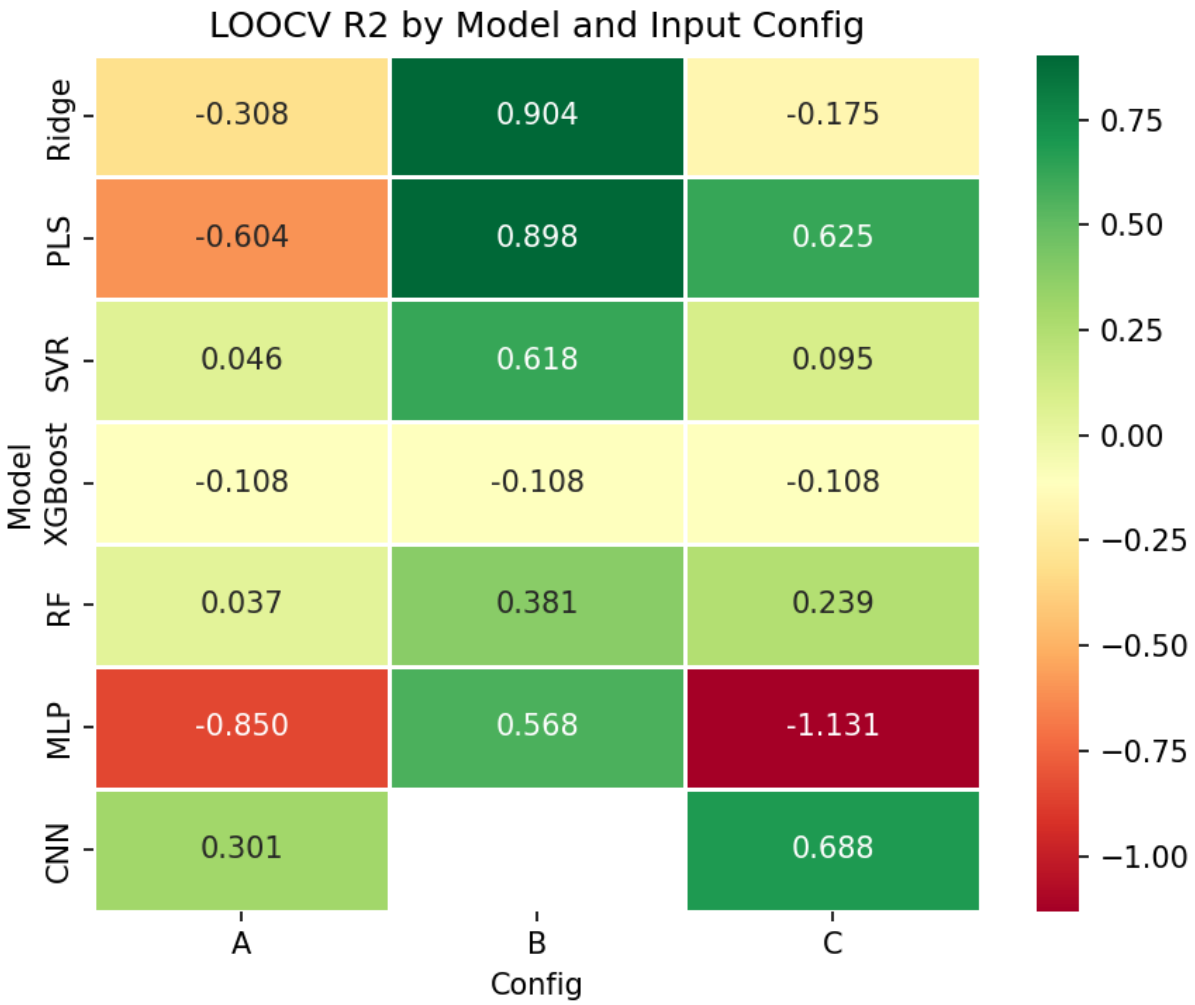
- **Discharge parameters:** frequency, pulse width, rise time, flow rate
- **Models tested:** Ridge, PLS, SVR, XGBoost, RF, MLP, CNN
- **Evaluation:** Leave-One-Out Cross-Validation (LOOCV) on 20 samples
- **Target:** H_2O_2 yield rate (range: 0.02–0.83)

Phase 1: PCA Dimensionality Reduction

- 701-channel OES spectrum reduced via PCA
- 11 principal components retained (95% cumulative variance)
- Severe curse of dimensionality: p/n ratio too high for 20 samples



Phase 1: Results — R^2 Heatmap (7 Models $\times$ 3 Configs)



- **Best:** Ridge Config B achieved $R^2 = 0.90$ (discharge params

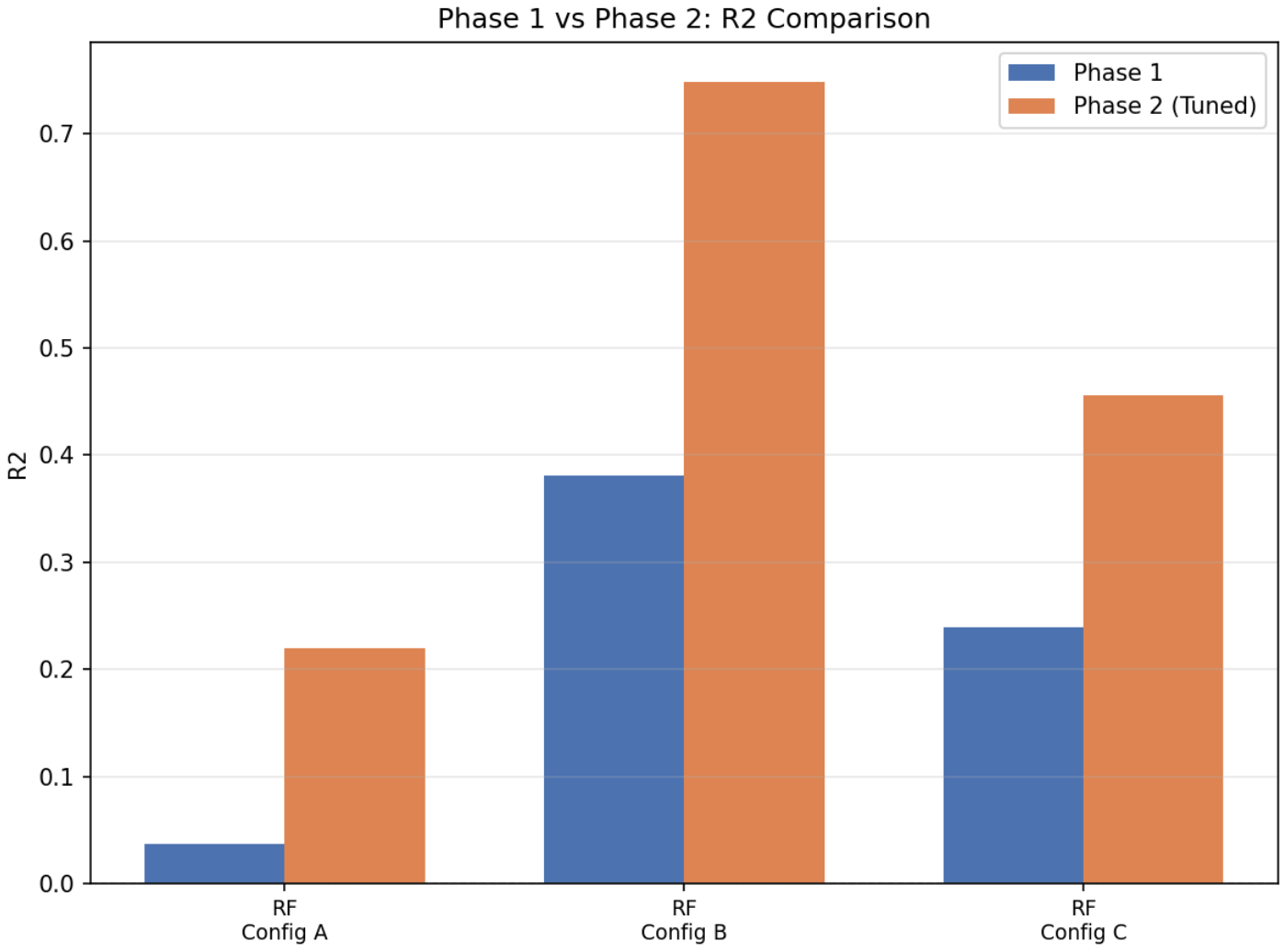
Phase 2: Bayesian Optimization

- Applied **Optuna** (TPE sampler) with nested LOOCV
- Tuned RF, MLP, and CNN: 100–200 trials per model-config pair
- Goal: recover nonlinear models from poor Phase 1 defaults

Key improvements after tuning:

Model	**Config**	**R^2 (P1)**	**R^2 (P2)**	**ΔR^2**
MLP	C	-1.13	0.37	$+1.50$
MLP	A	-0.85	0.37	$+1.22$
RF	B	0.38	0.75	$+0.37$
MLP	B	0.57	0.86	$+0.29$
CNN	C	0.69	0.77	$+0.09$

Phase 2: Results — Phase 1 vs Phase 2 Comparison



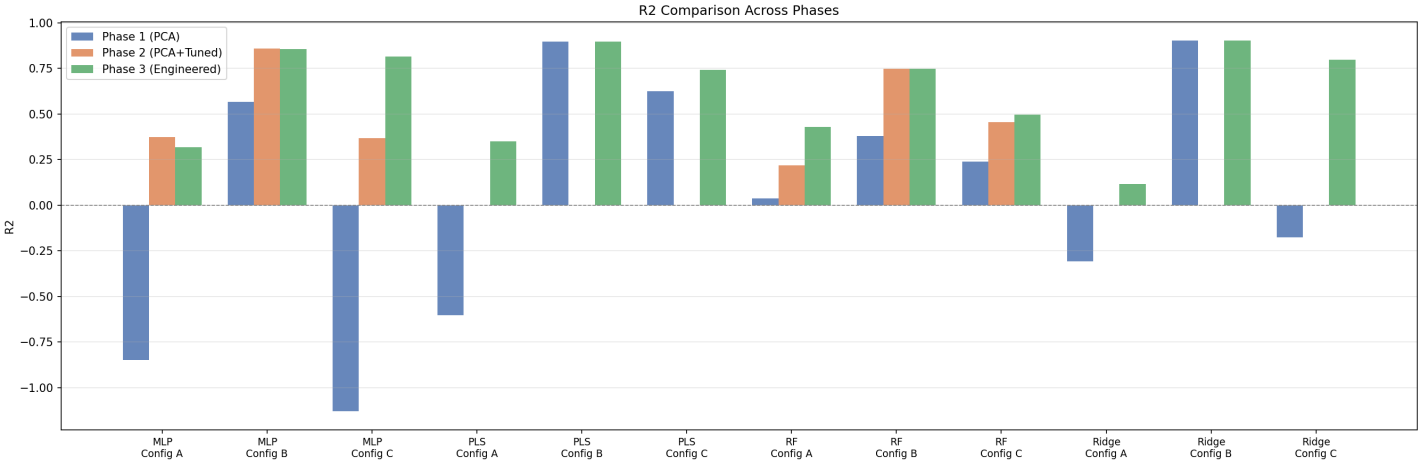
- Tuning universally beneficial, but Config B still dominant
- OES features via PCA remained weak even with optimized

Phase 3: 13 Physics-Based OES Features

Replaced blind PCA with features grounded in plasma chemistry:

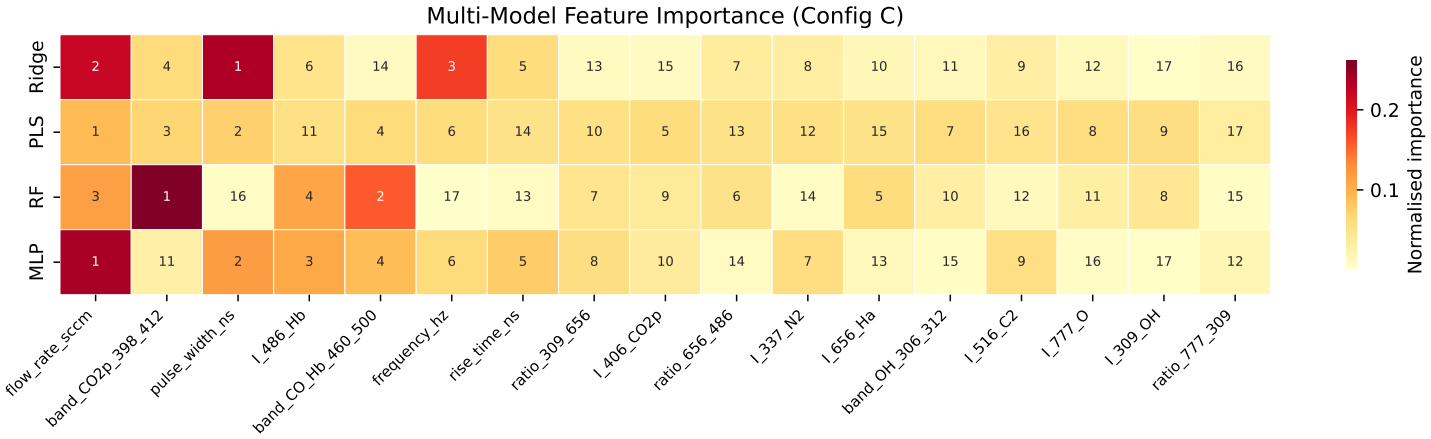
ID	Feature	Type	Species	Role
F1	I(309 nm)	Intensity	OH	Direct H_2O_2 precursor
F2	I(777 nm)	Intensity	O atom	CO_2 dissociation product
F3	I(656 nm)	Intensity	Hα	H_2O dissociation degree
F4	I(486 nm)	Intensity	Hβ	Electron diagnostics
F5	I(337 nm)	Intensity	N_2 SPS	High-energy e^- marker
F6	I(406 nm)	Intensity	CO_2^+	CO_2 ionization degree
F7	I(516 nm)	Intensity	C_2	Deep decomposition
F8	$\int$(306–312)	Band integral	OH 0-0 band	Robust OH emission
F9	$\int$(398–412)	Band integral	CO_2^+ FDB	Aggregate ionization
F10	$\int$(460–500)	Band integral	CO + Hβ	CO + H_2O dissociation
F11	I(309)/I(656)	Ratio	OH/Hα	OH availability
F12	I(777)/I(309)	Ratio	O/OH	Radical pool balance
F13	I(656)/I(486)	Ratio	Hα/Hβ	Balmer decrement

Phase 3: Results — Cross-Phase Comparison



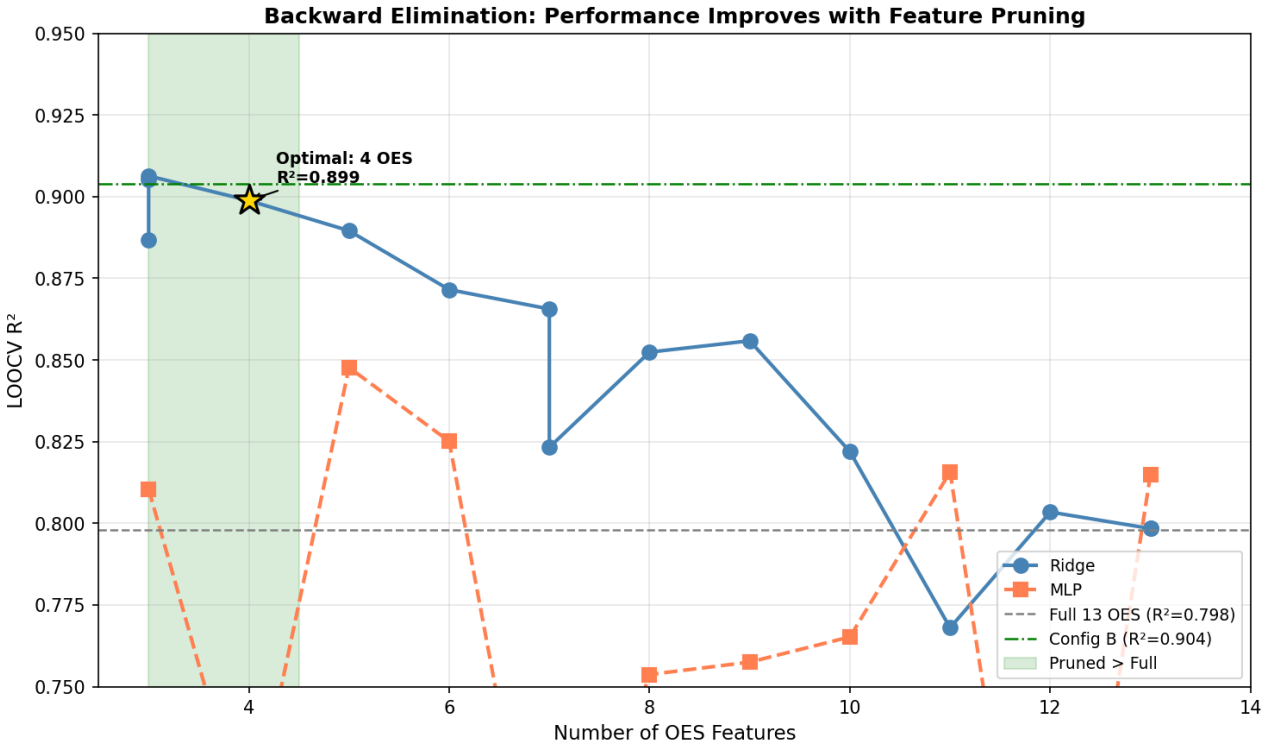
- **Ridge Config C:** R^2 improved from -0.17 to 0.80 ($+0.97$)
- **MLP Config C:** R^2 improved from -1.13 to 0.82 ($+1.95$)
- Domain knowledge dramatically unlocked OES predictive power

Phase 4: Multi-Model Feature Importance Consensus



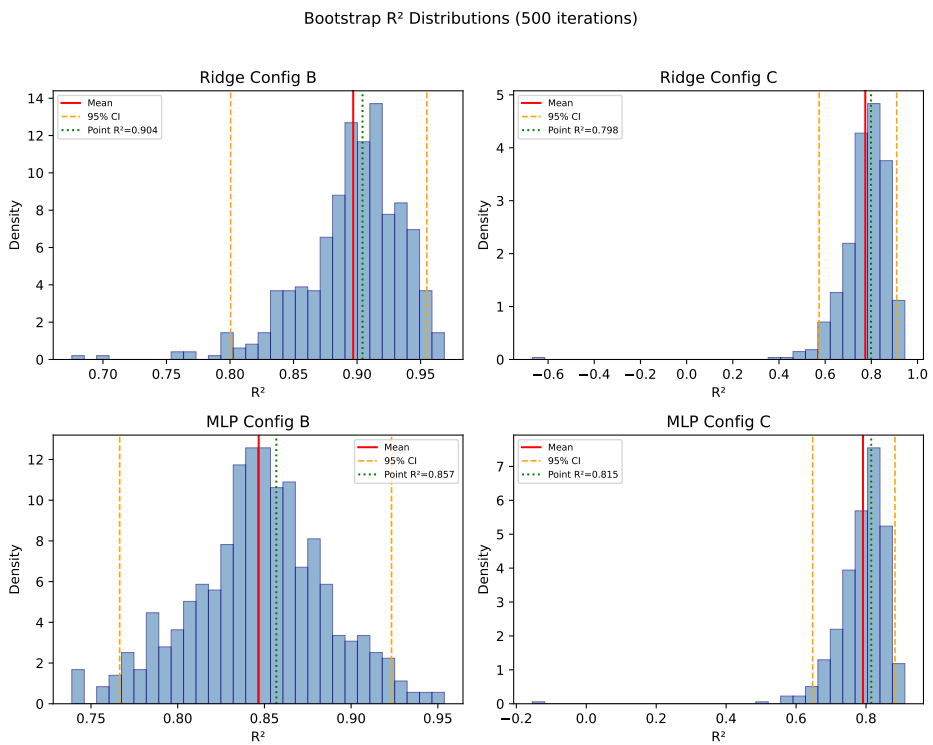
- Ranked 17 features across Ridge, PLS, RF, and MLP (SHAP)
- **Top features:** flow rate, CO_2^+ band integral, pulse width

Phase 4: Backward Elimination — Ablation Trajectory



- Removing redundant OES features **improves** R^2 (collinearity reduction)
- Optimal: **4 OES + 4 discharge params** $\rightarrow$ $R^2 = 0.899$
- Category ablation: 3 ratios alone + 4 params $\rightarrow$ $R^2 = 0.920$

Phase 4: Statistical Validation



Bootstrap 95% CIs (500 resamples):

Model	**R^2 [95% CI]**
Ridge B	0.90 [0.80, 0.95]
Ridge C	0.80 [0.57, 0.91]
MLP C	0.82 [0.65, 0.88]
MLP B	0.86 [0.77, 0.92]

Permutation test (2000 iter.):

- Observed $R^2 = 0.920$
- $p < 0.0005$

Phase 4: Final Optimized Model

Property	**Value**
Model	Ridge Regression
Total features	7
OES features (3)	I(309)/I(656), I(777)/I(309), I(656)/I(486)
Discharge params (4)	frequency, pulse width, rise time, flow rate
R^2 (LOOCV)	**0.920**
Permutation *p*-value	< 0.0005

- 3 OES intensity ratios + 4 discharge parameters = **7 features total**
- Outperforms the full 17-feature model (R^2 = 0.798) by removing collinear noise
- Simple, interpretable, and statistically significant

Progress Summary: R^2 Evolution Across Phases

Phase	Best Model	Config	R^2	Key Advancement
1	Ridge	B	0.904	Baseline; PCA for OES failed
2	MLP	B	0.861	Bayesian hyperparameter tuning
3	MLP	C	0.815	Domain-knowledge OES features
4	Ridge	C (pruned)	**0.920**	Feature reduction + interpretability

Key takeaways:

- Domain knowledge + feature reduction outperformed complex models on small data
- OES features became useful only after physics-guided engineering (Phase 3)
- Final model: simple Ridge with 7 features achieves $R^2 = 0.920$ ($p < 0.0005$)

Reference I

Thank You !